Lab Program

1. Missionaries and Cannibals Problem

Aim

To solve the Missionaries and Cannibals problem using Python.

Algorithm

1. Start with 3 missionaries and 3 cannibals on the left side.
2. The boat carries 1 or 2 people.
3. Never allow cannibals to outnumber missionaries where missionaries are present.
4. Move people safely until everyone reaches the right side.

Pseudocode

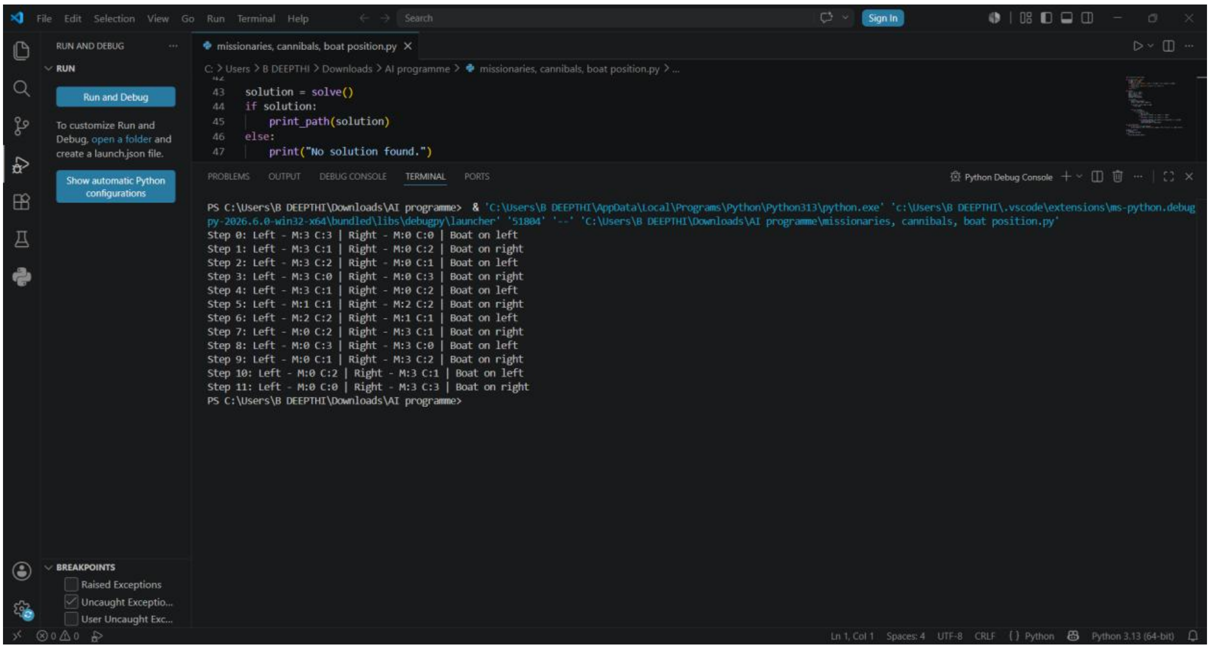
```
Start
Initial state = (3,3,1)
Goal state = (0,0,0)

While goal is not reached:
   Generate possible boat moves
   Check safe state
   Add new state
Print solution
Stop
```

Code

```
from collections import deque
start = (3, 3, 1)
goal = (0, 0, 0)
moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]
queue = deque([start])
visited = {start}
while queue:
   m, c, b = queue.popleft()
```

```
    print((m, c, b))
    if (m, c, b) == goal:
        break
    for x, y in moves:
        if b == 1:
            new = (m-x, c-y, 0)
        else:
            new = (m+x, c+y, 1)
        nm, nc, nb = new
        if 0 <= nm <= 3 and 0 <= nc <= 3:
            if (nm == 0 or nm >= nc) and (3-nm == 0 or 3-nm >= 3-nc):
                if new not in visited:
                    visited.add(new)
                    queue.append(new)
```

output :



Result:

The Program Missionaries and Cannibals Problem was excueted successfully

2. Vacuum Cleaner Problem

Aim

To solve the Vacuum Cleaner problem using Python.

Algorithm

1. Start with two rooms.
2. Check the current room.
3. If the room is dirty, clean it.
4. Move to the next room.
5. Repeat until all rooms are clean.

Pseudocode

```
Start
Set rooms as Dirty
For each room:
    If room is Dirty:
        Clean the room
Print final state
Stop
```

Code

```
def vacuum_cleaner(location, state):
    print("Vacuum is placed in Room", location)
    print("Room A:", state[0], " | Room B:", state[1])
    if location == "A":
        if state[0] == "Dirty":
            print("Room A is Dirty. Cleaning Room A...")
            state[0] = "Clean"
        else:
            print("Room A is already Clean.")
        location = "B"
        if state[1] == "Dirty":
            print("Room B is Dirty. Cleaning Room B...")
```

```
            state[1] = "Clean"
        else:
            print("Room B is already Clean.")
    elif location == "B":
        if state[1] == "Dirty":
            print("Room B is Dirty. Cleaning Room B...")
            state[1] = "Clean"
        else:
            print("Room B is already Clean.")
        location = "A"
        if state[0] == "Dirty":
            print("Room A is Dirty. Cleaning Room A...")
            state[0] = "Clean"
        else:
            print("Room A is already Clean.")
    print("\nFinal State: Room A:", state[0], "| Room B:", state[1])
vacuum_cleaner("A", ["Dirty", "Clean"])
```

output